

Why study development?

A young child with dark hair is swimming underwater, looking directly at the camera. The child is wearing a pink and green swimsuit. The background is a blue, rippling water surface.

T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

Research participation requirement

- Read the paragraph about this in the syllabus
- Start date is in a couple weeks
- More studies will be posted as the semester progresses
- You must arrive on time to studies
- Email rosemari@princeton.edu with questions

Research participation assignment

You must complete the Research Engagement Assignment in order to pass this course. Four hours of psychology experiment participation are assigned to any student in this course who has not already completed 8 hours of participation for other psychology courses in the past. Students will have access to sign up for experiments beginning September 18, 2025, and must complete the required number of sessions by December 4, 2025. See 'Research Engagement Assignment' document on Canvas for complete instructions.

This assignment reflects the psychology department's belief that experiencing research as a participant adds greatly to a student's understanding of course material, particularly to the student's understanding of how psychologists study behavior. Your participation not only furthers your education about the nature of psychological research; it also makes a substantial, critical contribution to psychological research at Princeton and to science in general. As an alternative to research participation, students may complete the Research Engagement Writing Assignment. Each paper is worth 1 hour of credit. Four papers would be required if you choose not to participate in any experiments. All questions should be directed to RoseMarie Stevenson (rosemari@princeton.edu).

Announcements

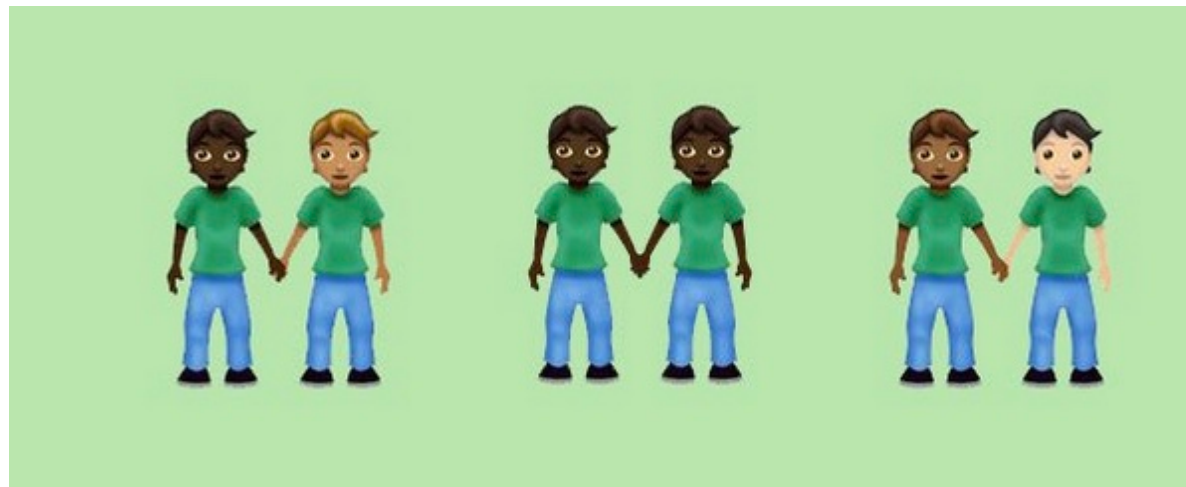
- First 'Multiple Choice Question' is due on Canvas before your precept next week
 - Create your own original question
 - Multiple-choice format only. Focus on content from lecture *or* precept.
 - Put some thought into it. Make it useful for your learning!
- Alternatively, you can write a 2-paragraph response to the week's material













The baby, assailed by eyes, ears, nose, skin,
and entrails at once,
feels it all as one great blooming, buzzing confusion.

-William James



Why study developmental psychology?

- Nothing is more important in a society than raising children. This field makes a difference.
- All we study in any other type of psychology is rooted in development.
- We can understand how learning happens. We can help.
- Children are weird.
- **It's an *entirely* different way of thinking. If (for example) you're a math major, this will feel truly distinct to you. You're in for a big treat.**

“Fast water”

Developmental themes

1. Why are babies so pathetic?

BACK the ONION

Study Reveals: Babies Are Stupid

LOS ANGELES--A surprising new study released Monday by UCLA's Institute For Child Development revealed that human babies, long thought by psychologists to be highly inquisitive and adaptable, are actually extraordinarily stupid.

The study, an 18-month battery of intelligence tests administered to over 3,500 babies, concluded categorically that babies are "so stupid, it's not even funny."

Sample Results:

Baby #2,678:



According to Institute president Molly Bentley, in an effort to determine infant survival instincts when attacked, the babies were prodded in an aggressive manner with a broken broom handle. Over 90 percent of them, when poked, failed to make even rudimentary attempts to defend themselves. The remaining 10 percent responded by vacating their bowels.



Above: Despite their relatively large cranial capacities, babies such as this one are so unintelligent that they are unable to distinguish colorful plastic squeak toys from food sources.

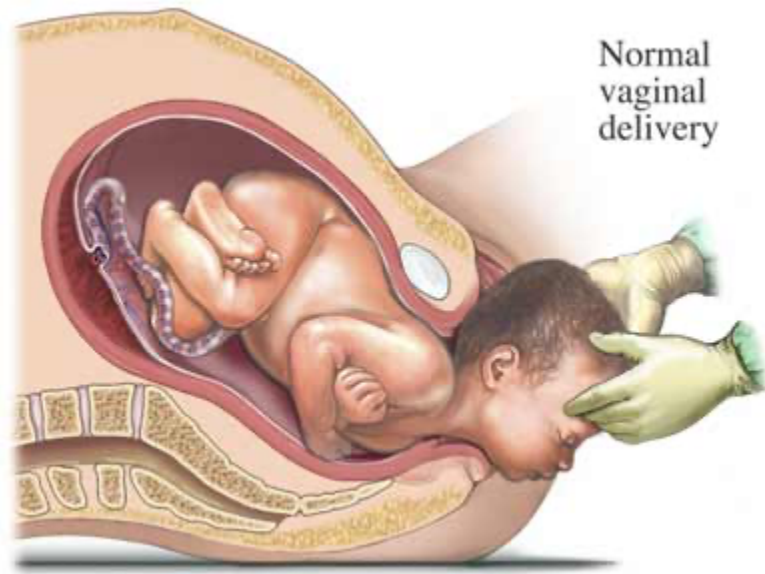
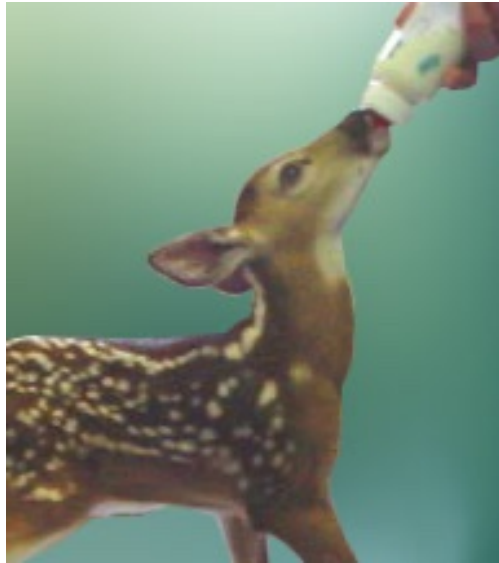
Baby #856:



**TEST
SITUATION:
PLACED IN
CENTER OF
LAKE ERIE IN
ROWBOAT
WITH
NAUTICAL MAP
TO SHORE
PROVIDED.**

**RESULT:
STILL ADRIFT
NEAR ST.
LAWRENCE
SEAWAY.**

1. Why are babies so pathetic?



1. Why are babies so pathetic?



Basic deficit in understanding of physical objects?

1. Why are babies so pathetic?



No awareness of radical scale changes?

2. Why are babies so smart?

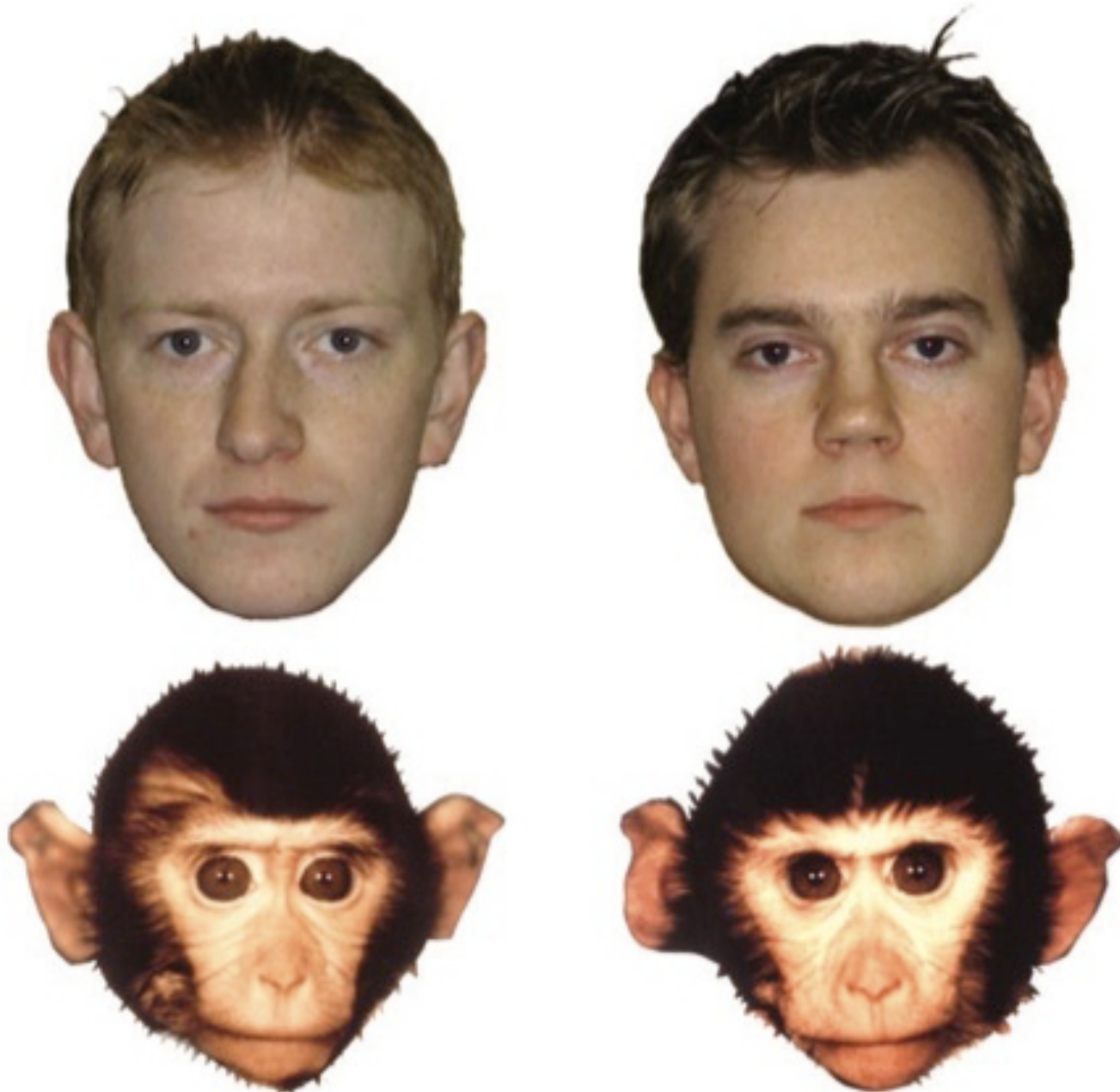
**This baby kinda knows
7,139 languages**



2. Why are babies so smart?



2. Why are babies so smart?



3. What makes us who we are?

Our genes?



Our families?



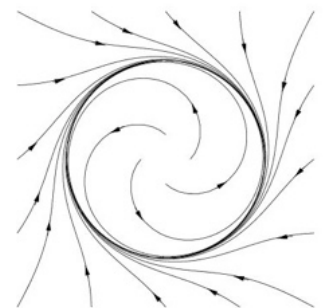
Our peers?



Our cultures?



The complex dynamics of our brains/ bodies/ environments?



3. What makes us who we are?

- Population-level risks associated with **low birthweight**
 - Newborn risks (breathing, temperature regulation, feeding, jaundice, infections)
 - Delayed growth
 - Developmental challenges (e.g., learning difficulties, ADHD)
 - Cerebral palsy
 - Vision/hearing problems
 - Behavioral and emotional difficulties
 - Adolescent/adult health (cardiovascular disease, Type 2 diabetes, kidney disease, pulmonary problems, mental health vulnerabilities)

3. What makes us who we are?



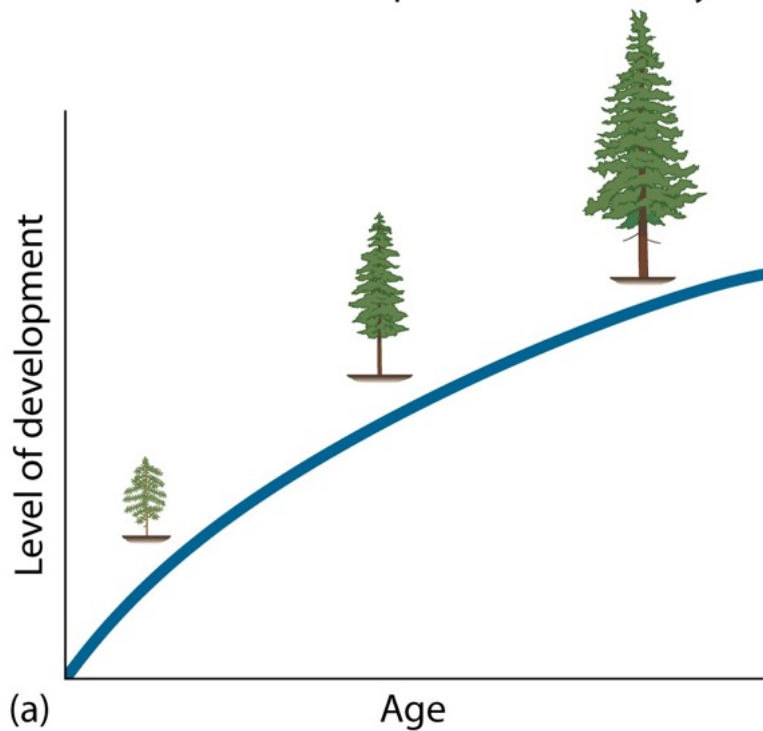
4. What changes over development? *and how does change happen?*



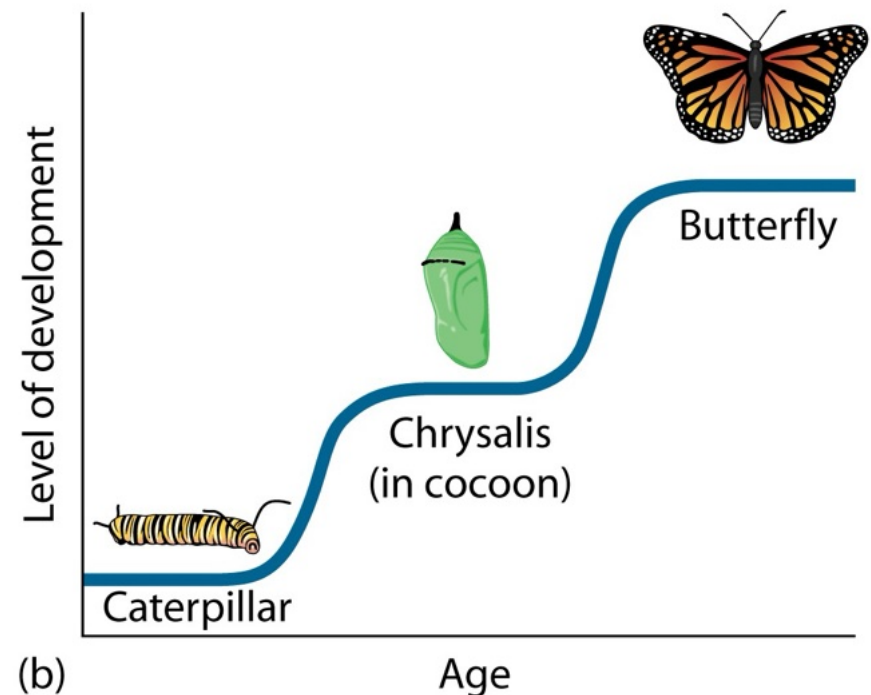
4. What changes over development? *and how does change happen?*

- Do we change continuously and incrementally across development, or do we make big shifts to qualitatively new behaviors?

Pine tree: Developmental continuity



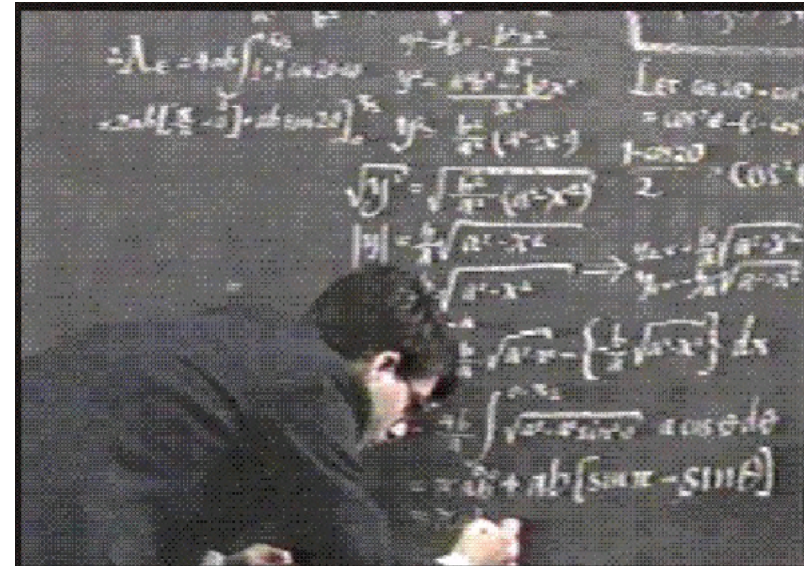
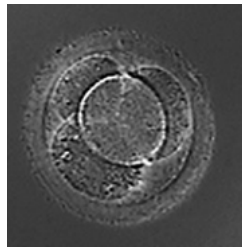
Butterfly: Developmental discontinuity



4. What changes over development? *and how does change happen?*



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5. How variable is early learning?



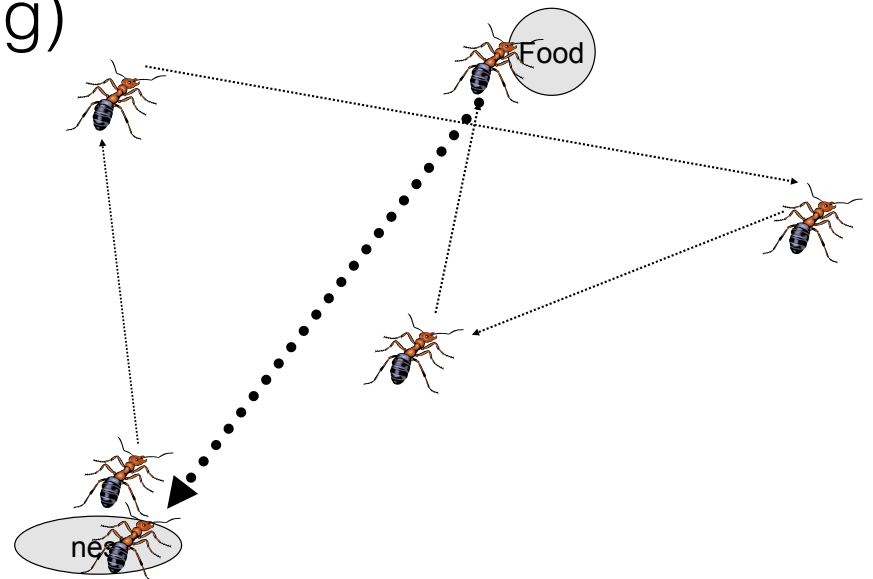
6. How do children shape their own development?

It's not just caregivers. Children are actively involved in their own development.

- Different kids in different situations elicit different responses from adults and other kids
- They have varied interests from infancy onwards, which leads to different expertise

7. How do nature and nurture shape development?

- **Nature:** our biological endowment
- Experience-independent change
- What counts as **nature**?
 - Genetic factors (chromosomal abnormalities, eye color)
 - Maturational timetables (growing teeth)
 - Hormonal changes (menstruation, birdsong)
 - Reflexes
 - Instincts (spider web, imprinting)



7. How do nature and nurture shape development?

- **Nurture**: our environment, including physical and social experience
- What counts as **nurture**?
 - Environmental factors (womb, native language, nutrition)
 - Learning (playing piano, attending school)
 - Culture
 - Effects of experience (food preferences, development of prejudice)
- Is nature **vs.** nurture the right way to think?

Nature x Nurture!



Nature x Nurture!



Nature x Nurture!



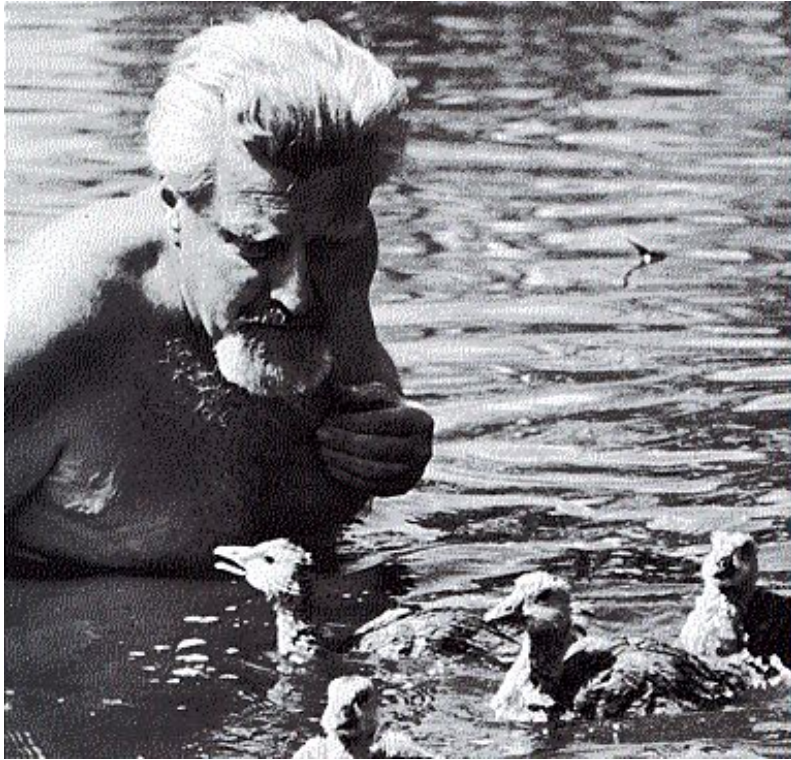
Nature and nurture shape the individual, but also shape each other. Nature influences how we experience nurture, and nurture influences the way nature expresses itself.

Thinking about nativism vs. empiricism

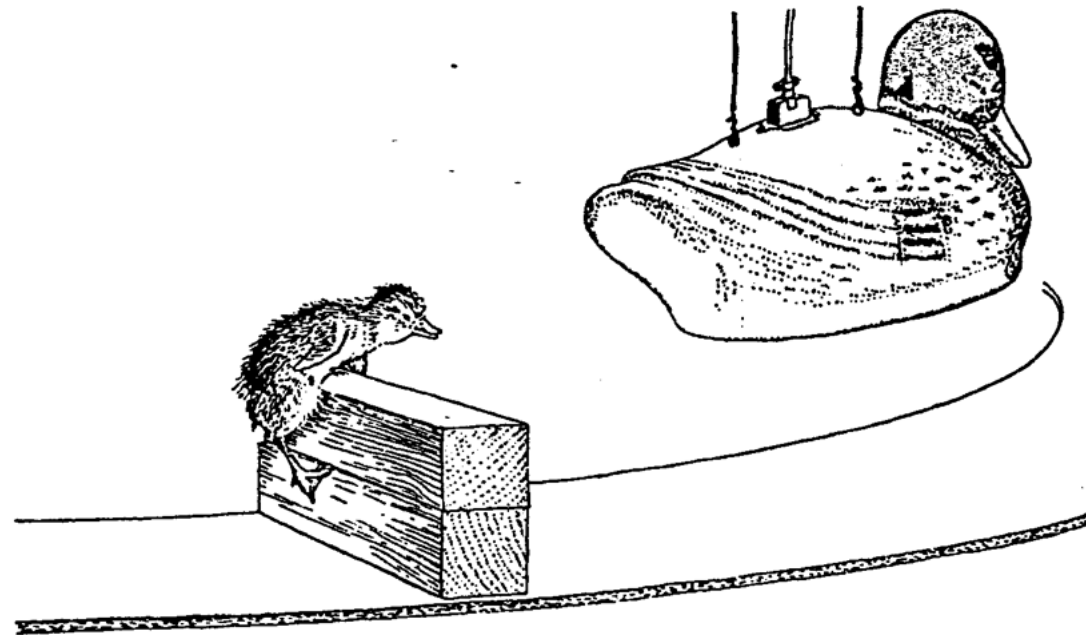
- Age of emergence
- Cross-cultural consistency
- Species comparisons



We *can* tease apart nature and nurture in some cases



Konrad Lorenz
1927-1989



DUCKLING SCALES AN OBSTACLE in the runway in an experiment to determine whether the effort it expends during imprinting is related to its score when it is tested.

But in many cases, it's extremely hard to disentangle



Kids with genes that correlate with school success also evoke more intellectual engagement from their parents (compared to siblings who don't have these genes)

Many more examples can be found in a 2025 book by Princeton sociology professor Dalton Conley, *The Social Genome*.

But in many cases, it's extremely hard to disentangle

Br. J. Nutr. (1980), **43**, 403

403

The effect of rehabilitation at different stages of development of rats marginally malnourished for ten to twelve generations

**BY R. J. C. STEWART, HILDA SHEPPARD,
R. PREECE AND J. C. WATERLOW**

*Clinical Nutrition and Metabolism Unit, London School of Hygiene
and Tropical Medicine, Keppel Street, London WC1E 7HT*

But in many cases, it's extremely hard to disentangle

Archival Report

Biological
Psychiatry

Holocaust Exposure Induced Intergenerational Effects on *FKBP5* Methylation

Rachel Yehuda, Nikolaos P. Daskalakis, Linda M. Bierer, Heather N. Bader, Torsten Klengel, Florian Holsboer, and Elisabeth B. Binder

ABSTRACT

BACKGROUND: The involvement of epigenetic mechanisms in intergenerational transmission of stress effects has been demonstrated in animals but not in humans.

METHODS: Cytosine methylation within the gene encoding for FK506 binding protein 5 (*FKBP5*) was measured in Holocaust survivors ($n = 32$), their adult offspring ($n = 22$), and demographically comparable parent ($n = 8$) and offspring ($n = 9$) control subjects, respectively. Cytosine-phosphate-guanine sites for analysis were chosen based on their spatial proximity to the intron 7 glucocorticoid response elements.

RESULTS: Holocaust exposure had an effect on *FKBP5* methylation that was observed in exposed parents as well in their offspring. These effects were observed at bin 3/site 6. Interestingly, in Holocaust survivors, methylation at this site was higher in comparison with control subjects, whereas in Holocaust offspring, methylation was lower. Methylation levels for exposed parents and their offspring were significantly correlated. In contrast to the findings at bin 3/site 6, offspring methylation at bin 2/sites 3 to 5 was associated with childhood physical and sexual abuse in interaction with an *FKBP5* risk allele previously associated with vulnerability to psychological consequences of childhood adversity. The findings suggest the possibility of site specificity to environmental influences, as sites in bins 3 and 2 were differentially associated with parental trauma and the offspring's own childhood trauma, respectively. *FKBP5* methylation averaged across the three bins examined was associated with wake-up cortisol levels, indicating functional relevance of the methylation measures.

CONCLUSIONS: This is the first demonstration of an association of preconception parental trauma with epigenetic alterations that is evident in both exposed parent and offspring, providing potential insight into how severe psychophysiological trauma can have intergenerational effects.

To understand development and learning, you have to get clever...

